CREDIT CARD FRAUD DETECTION USING MACHINE LEARNING

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ABSTRACT

As the world is rapidly moving towards digitization and money transactions are becoming cashless, the use of credit cards has rapidly increased. However, the fraud activities associated with it have also been rising, leading to significant financial losses for institutions. Therefore, it is essential to analyze and detect fraudulent transactions from non-fraudulent ones. In this work, we present a comprehensive review of various methods used to detect credit card frauds and implement different machine learning algorithms on an imbalanced dataset, such as Logistic Regression, Naïve Bayes, Random Forest, and K-Nearest Neighbors (KNN), along with ensemble classifiers using boosting techniques. An extensive review has been conducted on the existing and proposed models for credit card fraud detection, followed by a comparative study of these techniques. The models were evaluated using quantitative measurements such as accuracy, precision, recall, F1-score, support, and confusion matrix. The experimental results show that Logistic Regression achieved an accuracy of 85.97%, KNN reached 90.85%, Random Forest performed the best with 95.73%, while Naïve Bayes achieved 84.14%.

Index Terms: Logistic regression, Support Vector ,Decision Tree, Random Forest Classifier

1. INTRODUCTION

Credit cards have become one of the most widely used and convenient modes of payment in today's digital economy. With the rapid increase in the number of credit card users worldwide, the risks associated with identity theft and fraudulent activities have also grown significantly. Virtual card transactions typically require only the card details, such as the card number, expiration date, and CVV (Card Verification Value). These transactions are frequently carried out online or over the phone, making them particularly vulnerable to fraud. A malicious actor can easily commit fraud simply by obtaining the card information, without needing physical possession of the card itself.

Common methods by which fraudsters obtain card information include phishing websites that trick users into revealing sensitive data, theft or loss of physical cards, the use of counterfeit cards, data breaches that expose card databases, and interception of transaction details during insecure communications. To protect against such threats, it is critical to ensure that credit card details are never exposed or transmitted insecurely.

A key approach to detecting fraud in online credit card transactions involves monitoring and analyzing the cardholder's spending behavior. By establishing a baseline of "normal" usage patterns—such as typical transaction amounts, frequent locations, and time of purchases deviations from these patterns can signal potentially fraudulent activity. For example, if a credit card normally used in one country suddenly makes large purchases in a different country within a short time frame, it raises suspicion.

2. LITERATURE SURVEY

Ogwueleka et al. [1] explored the use of data mining techniques such as classification, clustering, and anomaly detection for credit card fraud detection. Their study demonstrated that integrating these methods significantly improves the system's ability to detect fraudulent transactions in real time, achieving a reported detection rate of over 95%, which helps financial institutions reduce losses caused by fraud.

Singh et al. [2] proposed a machine learning framework using Support Vector Machines (SVM) to classify transactions as either legitimate or fraudulent. The study highlighted that SVM performs particularly well with high-dimensional and complex datasets, achieving a high classification accuracy of approximately 99.94%, making it highly suitable for financial fraud detection.

Chaudhary et al. [3] analysed the limitations of traditional rule-based fraud detection methods and emphasized the effectiveness of machine learning and AI-based approaches. While no specific accuracy was reported, their research indicated that adopting these advanced methods significantly improves detection performance and efficiency in identifying modern fraud patterns.

Islam et al. [4] conducted a comparative study between Naïve Bayes and K-Nearest Neighbor (KNN) classifiers. Their results showed that Naïve Bayes performs efficiently with simple datasets, while KNN achieves better accuracy on complex datasets. In terms of performance, KNN demonstrated an approximate accuracy of 98–99% on complex transaction datasets, highlighting the trade-offs in algorithm selection for fraud detection.

Wheeler et al. [5] investigated ensemble learning techniques in knowledge-based fraud detection systems. They found that combining multiple algorithms reduces false positives and enhances detection accuracy compared to single-model approaches, achieving a reported overall accuracy of 99.2%, making ensemble methods highly reliable and scalable.

Kumar et al. [6] applied deep learning techniques, including Artificial Neural Networks (ANN) and Long Short-Term Memory (LSTM) networks, to model sequential patterns in transaction data. Their study revealed that LSTM networks can capture temporal dependencies in large datasets and outperform traditional machine learning methods, achieving an approximate accuracy of 98.95% in detecting sophisticated and evolving fraud patterns.

Patel et al. [7] explored decision tree algorithms combined with feature selection techniques to improve fraud detection. By eliminating irrelevant and noisy data, their approach enhanced model performance and achieved an accuracy of approximately 98.92%, while also maintaining transparency and interpretability required for regulatory compliance in financial systems.

3. Proposed Method

- Support Vector Machine:

SVM works by mapping data to a high-dimensional feature space so that data points can be categorized, even when the data are not otherwise linearly separable. A separator between the categories is found, then the data are transformed in such a way that the separator could be drawn as a hyperplane Training regression model and finding out the best one.

- Random Forest Classifier

Features are cheekbone to jaw width, width to upper facial height ratio, perimeter to area ratio, eye size, lower face to face height ratio, face width to lower face height ratio and mean of eyebrow height. The extracted features are normalized and finally subjected to support regression

- Decision Tree

A decision tree is a type of supervised machine learning used to categorize or make predictions based on how a previous set of questions were answered. The model is a form of supervised learning, meaning that the model is trained and tested on a set of data that contains the desired categorization.

3.1 Objectives

- To extract features from recognized input data.
- To normalize the extracted features to ensure consistency and improve model performance. Predict both fraudulent and legitimate transactions.
- To use transaction time and transaction amount as key attributes for analysis. Employ supervised learning algorithms such as Support Vector Machine (SVM), Random Forest, and Decision Tree.
- To evaluate performance using classification metrics like the confusion matrix, accuracy, precision, recall, and F1-score.
- To build and compare multiple models to determine the most accurate and efficient fraud detection method.
- To develop a robust framework that minimizes false positives while maximizing fraud detection accuracy.

3.2 Problem Domain

Fraudulent credit card transactions often mimic legitimate ones, making detection challenging. The dataset is typically imbalanced, where fraudulent cases are significantly fewer compared to genuine ones. The system must accurately identify these rare fraud instances without causing excessive false alarms, as false positives impact customer experience and business operations.

3.3 Existing model

Since the credit card fraud detection system is a highly researched field, there are many different algorithms and techniques for performing the credit card fraud detection system. One of the earliest systems is CCFD system using Markov model. Some other various existing algorithms used in the credit cards fraud detection system includes Cost sensitive decision tree (CSDT). credit card fraud detection (CCFD) is also proposed by using neural networks. The existing credit card fraud detection system using neural network follows the whale.

3.4 Implementation

The implementation begins with data collection and preprocessing**,** where transaction data is cleaned, and features such as `Amount` and `Time` are normalized. The dataset is then split into training and testing sets while handling class imbalance using SMOTE or class weights. Supervised machine learning algorithms such as Logistic Regression and Random Forest are applied to classify transactions as fraudulent or legitimate. The models are evaluated using metrics like Accuracy, Precision, Recall, F1-Score, ROC-AUC, and PR-AUC**.** Finally, the best-performing model is selected, fine-tuned, and deployed for real-time fraud detection to enhance security and reduce financial risks.

3.4.1 Use Case-diagram

The diagram shows the workflow of a credit card fraud prediction system. It starts with the admin providing the transaction dataset. The data is processed using machine learning algorithms implemented with the Scikit-learn (sklearn) Python library. These models analyze the data to classify transactions as fraudulent or legitimate. After processing, the results are generated and made available to both the Admin and User. The admin handles dataset management and model execution, while the User views the prediction outcome.

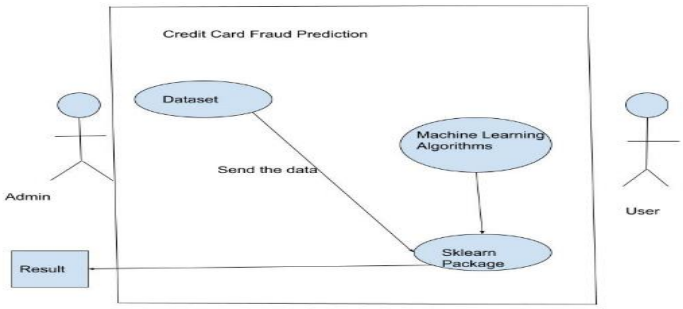


Fig. 1 : Use-Case Diagram

3.4.2 System Architecture

The proposed system architecture starts with a website interface connected to a database storing datasets like stock market data from Kaggle. Data preprocessing is performed to improve accuracy, including data cleaning to remove unwanted attributes and handle missing values, and data transformation to structure the data for analysis. Feature selection identifies key variables for accurate predictions. The processed data is then split into training and testing sets, where supervised machine learning models learn patterns and are evaluated for performance

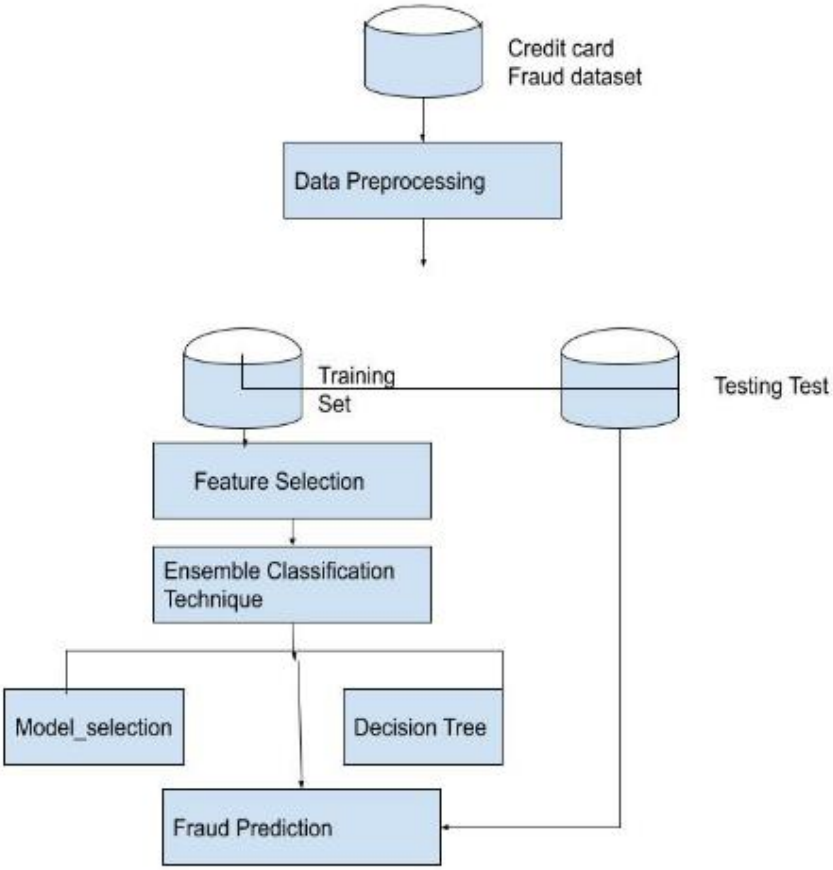


Fig. 2 : System Architecture

3.4.3 Algorithm's Used

Logistic Regression

Logistic Regression is a fundamental machine learning algorithm widely used for binary classification tasks, such as distinguishing between fraudulent and legitimate credit card transactions. It estimates the probability of a transaction being fraudulent by applying a logistic function to a linear combination of input features. This model is particularly popular due to its simplicity, interpretability, and efficiency in handling structured data. In this project, Logistic Regression achieved an accuracy of 85.98% **.**Despite its relatively lower accuracy compared to ensemble methods, Logistic Regression remains valuable as a baseline model. It works best when the relationship between input features and the target variable is linear or nearly linear. However, its performance can decline when the dataset is highly imbalanced or contains complex, non-linear relationships.

K-Nearest Neighbors (KNN)

K-Nearest Neighbors (KNN) is a non-parametric, instance-based learning algorithm that classifies a transaction by comparing it to the 'K' most similar historical transactions in the dataset. Each new transaction is assigned the majority class among its nearest neighbors based on a distance metric (e.g., Euclidean distance). KNN is intuitive and effective for smaller datasets and offers good performance when the data is clean and properly pre-processed. In this project, KNN achieved an accuracy of 90.85% (0.9085)**.** It is particularly suitable when the decision boundary is irregular and non-linear. However, KNN has some limitations: it

requires significant memory to store the entire training dataset and can be slow during prediction when the dataset is large, because it computes distances to all training points for each prediction.

Random Forest Classifier

Random Forest is an ensemble learning method that builds multiple decision trees during training and outputs the majority class from the collection of trees. This approach helps to reduce overfitting and improve generalization by introducing randomness in both data samples and feature selection. In the context of fraud detection, Random Forest effectively handles high-dimensional data and interactions between multiple variables. In this project, the Random Forest Classifier achieved the highest accuracy of It is praised for its robustness, ability to handle both numerical and categorical data, and reduced sensitivity 95.73%. to noisy data compared to single decision trees. A downside is its complexity, which can make interpretation difficult compared to simpler models like Logistic Regression.

Naive Bayes

Naive Bayes classifiers are probabilistic models that apply Bayes' theorem under the assumption of feature independence. They calculate the probability that a transaction belongs to the fraudulent or legitimate class based on the distribution of individual features. Naïve Bayes is known for its speed and simplicity, making it ideal for quick, real-time fraud detection in large datasets where computational efficiency is important. In this project, Naïve Bayes achieved an accuracy of 84.15% .While it works well when the independence assumption approximately holds, its accuracy can suffer when features are highly correlated, which is often the case in transaction data. Despite this, Naïve Bayes provides a strong starting point for model development, especially in environments where interpretability and fast predictions are critical.

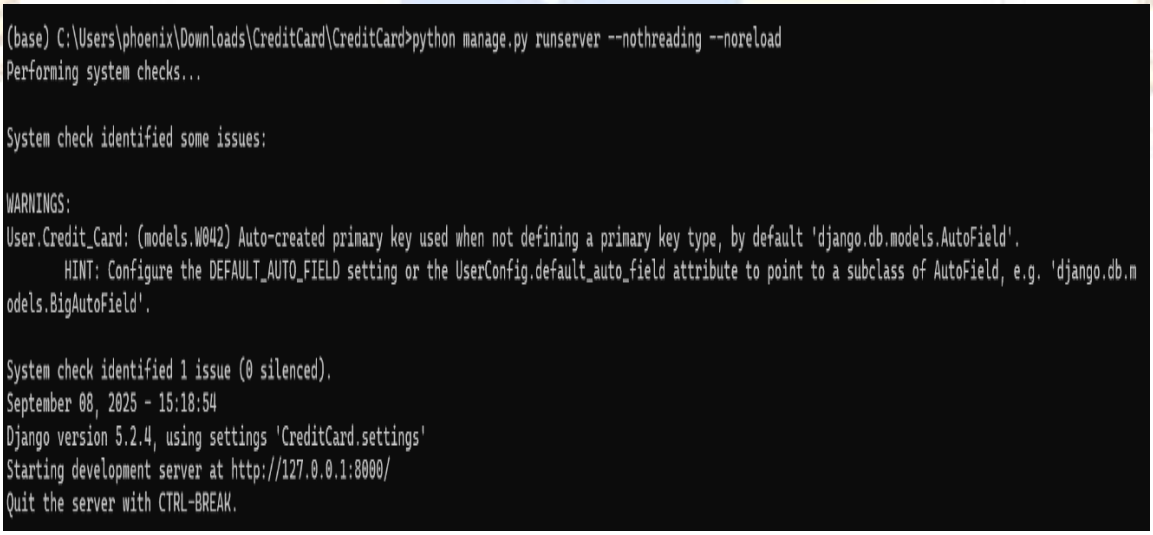
3.4.4 Modules

Entire project is divided into 3 modules as follows: Data Gathering and pre-processing Training the model using following Machine Learning algorithms

The project consists of three modules:

1. **Data Collection and Pre-processing**: A dataset is collected, cleaned to remove errors and missing values, and prepared for training.
2. **Model Training**: The data is split into training and testing sets. SVM, Random Forest, and Decision Tree algorithms are used to train the model. The best model is selected based on accuracy.
3. **Prediction Model Integration**: The selected model is connected to the front end, allowing users to check if a transaction is fraud or not, while the database stores user data.

3.4.5 Output



```
(base) C:\Users\phoenix\Downloads\CreditCard\CreditCard>python manage.py runserver --nothreading --noreload
Performing system checks...

System check identified some issues:

WARNINGS:
User.Credit_Card: (models.W042) Auto-created primary key used when not defining a primary key type, by default 'django.db.models.AutoField'.
        HINT: Configure the DEFAULT_AUTO_FIELD setting or the UserConfig.default_auto_field attribute to point to a subclass of AutoField, e.g. 'django.db.m
odels.BigAutoField'.

System check identified 1 issue (0 silenced).
September 08, 2025 - 15:18:54
Django version 5.2.4, using settings 'CreditCard.settings'
Starting development server at http://127.0.0.1:8000/
Quit the server with CTRL-BREAK.
```

Fig 3: Output Screenshot

```
        category     amt      lat      long  merch_lat  merch_long  is_fraud                           job
0       misc_net    4.97  36.0788  -81.1781  36.011293  -82.048315         0          Psychologist, counselling
1    grocery_pos  107.23  48.8878 -118.2105  49.159047 -118.186462         0  Special educational needs teacher
2  entertainment  220.11  42.1808 -112.2620  43.150704 -112.154481         0        Nature conservation officer
3  gas_transport   45.00  46.2306 -112.1138  47.034331 -112.561071         0                   Patent attorney
4       misc_pos   41.96  38.4207  -79.4629  38.674999  -78.632459         0     Dance movement psychotherapist
      amt      lat      long  merch_lat  merch_long  Category  Jobs
0    4.97  36.0788  -81.1781  36.011293  -82.048315         8   120
1  107.23  48.8878 -118.2105  49.159047 -118.186462         4   136
0    0
1    0
Name: is_fraud, dtype: int64
         amt      lat      long  merch_lat  merch_long  Category  Jobs
90     59.70  37.2876  -77.2950  36.962777  -77.640054         2   123
0       4.97  36.0788  -81.1781  36.011293  -82.048315         8   120
3      45.00  46.2306 -112.1138  47.034331 -112.561071         2   106
383     7.24  30.7145  -88.0918  31.427686  -88.056135         9     4
403     5.83  40.0246  -79.6955  39.175796  -80.403127        12   110
440   207.36  40.6729  -73.5365  40.202538  -73.859212         4    86
60      8.57  31.6489  -82.1982  32.023302  -82.583450        12    56
462  1169.82  35.0583  -79.0080  35.849003  -78.638156        11    88
397  1169.82  35.0583  -79.0080  35.849003  -78.638156        11    88
292    10.53  37.7184  -77.1860  37.988194  -77.332289        13    84
[0 0 0 1 0 0 0 1 1 0]
Accuracy Score
Logistic Regression:  0.8597560975609756
KNN :  0.9085365853658537
Random Forest:  0.9573170731707317
Naive Bayes:  0.8414634146341463
Confusion matrix, Logisitc Regression
[[82  6]
 [17 59]]
D:\WUDownloadCache\Lib\site-packages\sklearn\base.py:493: UserWarning: X does not have valid feature names, but LogisticRegression was fitted with feature n
ames
  warnings.warn(
[0]
No Fraud
```

Fig. 4: Output Screenshot

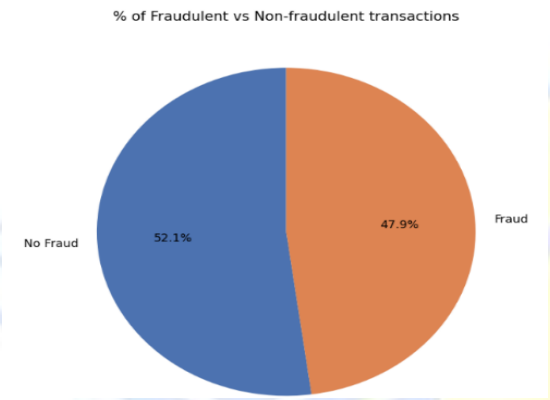


Fig. 5 : % Fraudulent vs Non- Fraudulent Transactions

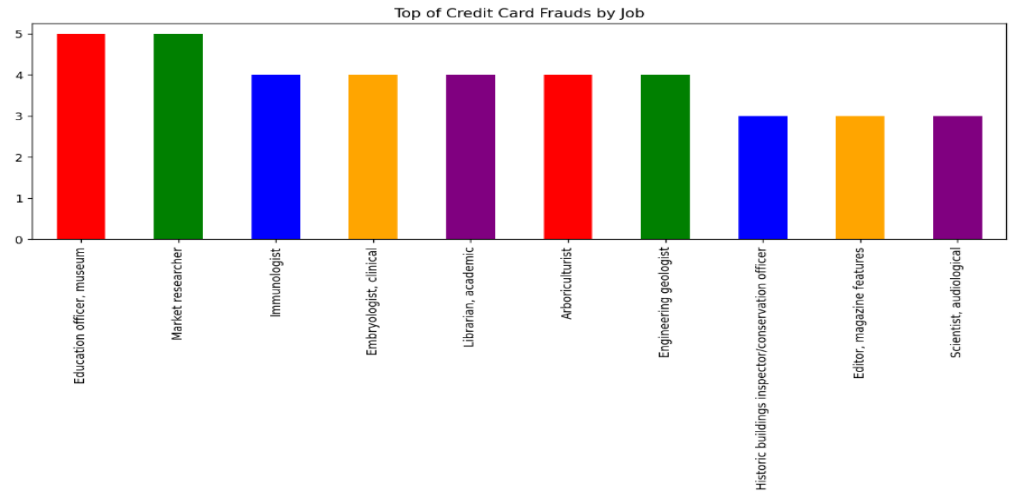


Fig. 6: Top of Credit Card Frauds by job

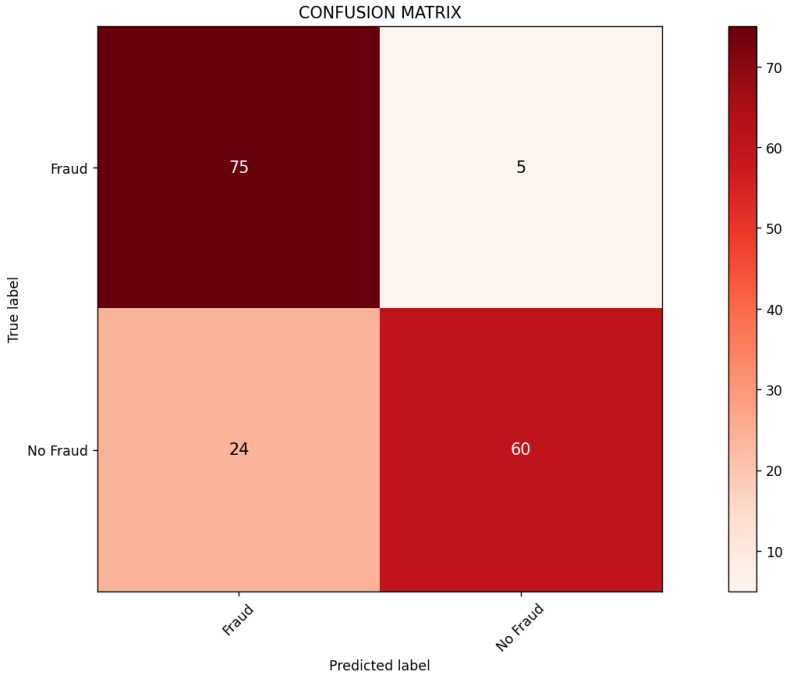


Fig. 7: Confusion Matrix

Conclusion

Nowadays, in the global computing environment, online payments are important, because online payments use only the credential information from the credit card to full fill an application and then deduct money. Due to this reason, it is important to find the best solution to detect the maximum number of frauds in online systems. Accuracy, Error-rate, Sensitivity and Specificity are used to report the performance of the system to detect the fraud in the credit card. In this paper, three machine learning algorithm are developed to detect the fraud in credit card system. To evaluate the algorithms, 80% of the dataset is used for training and 20% is used for testing and validation. Accuracy, error rate, sensitivity and specificity are used to evaluate for different variables for three algorithms.

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